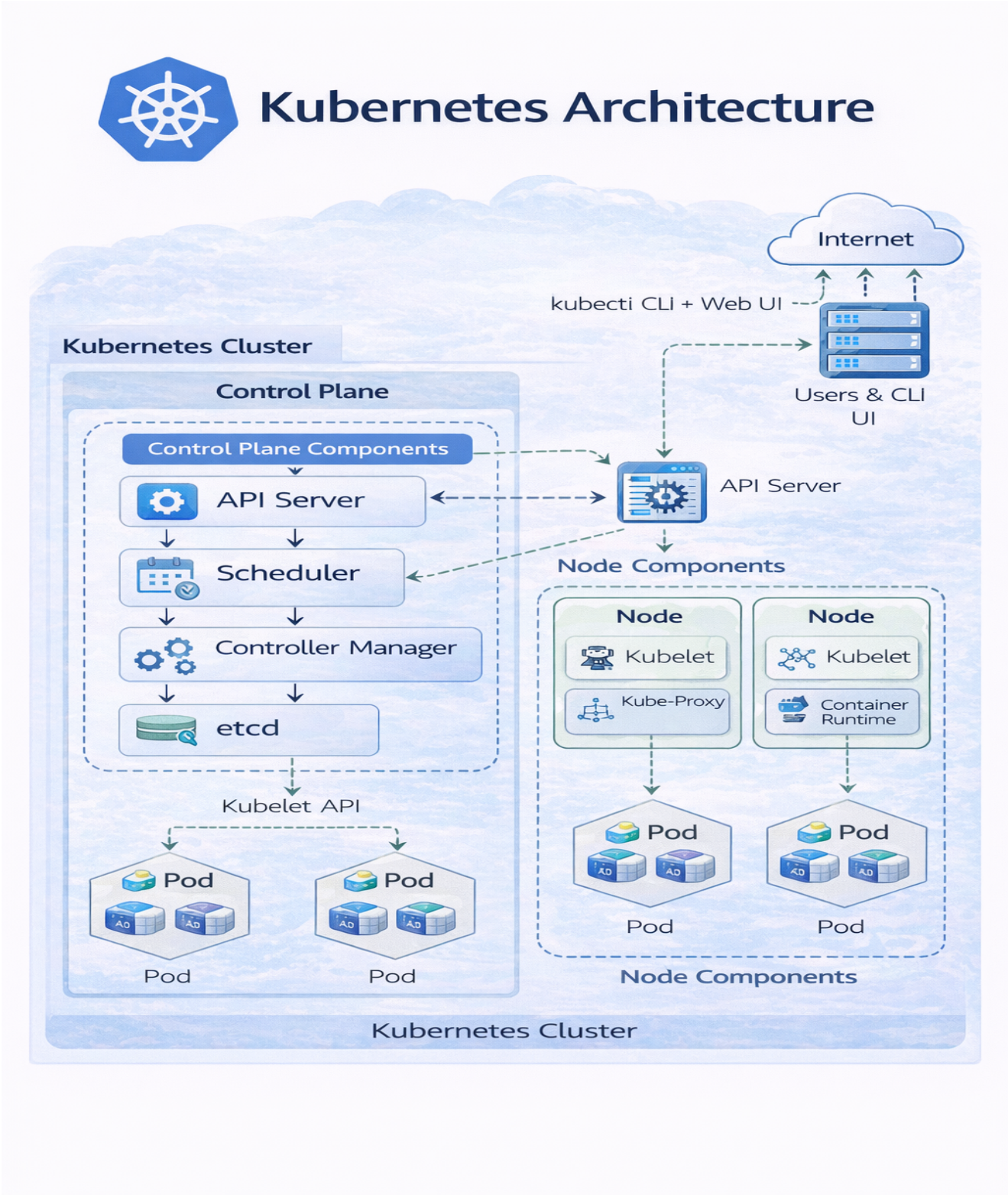
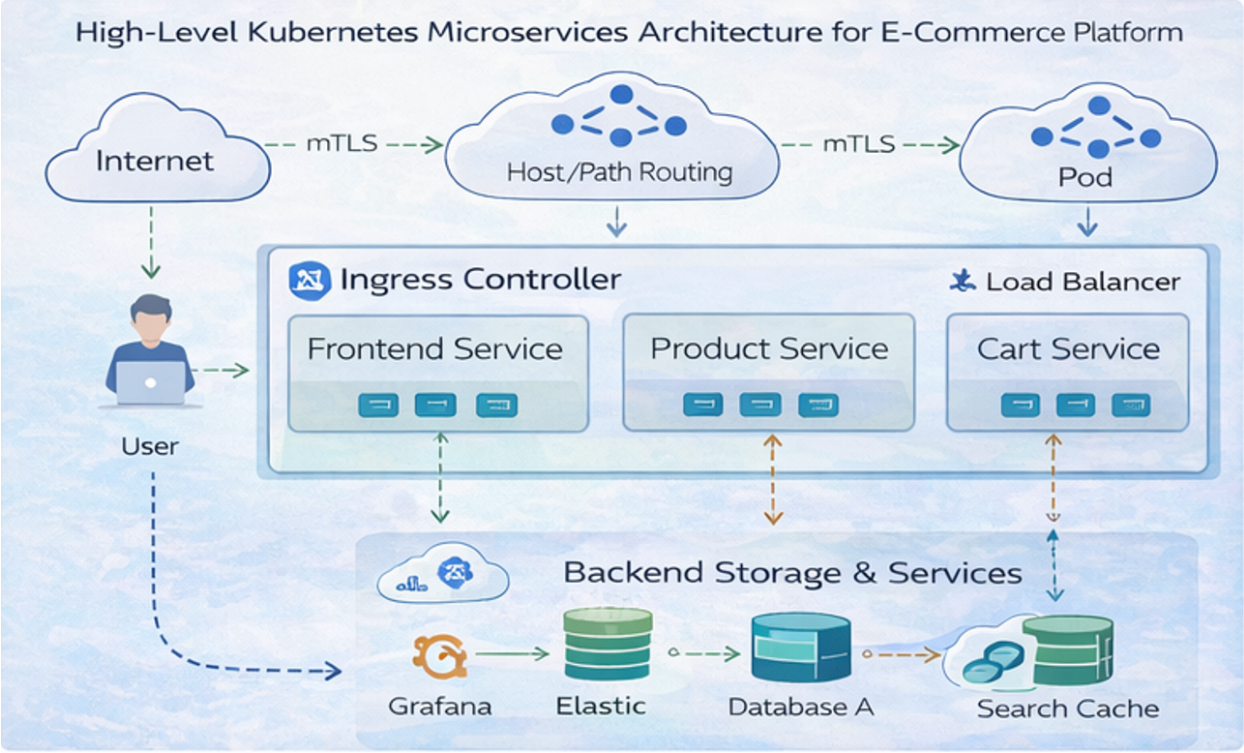
Kubernetes Interview Questions & Answers



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High-Level Kubernetes System Design Diagram

Below is a high-level production-ready Kubernetes microservices architecture illustrating ingress routing, services, pods, backend storage, observability, and external traffic flow.



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1) What is Kubernetes?

Kubernetes is a container orchestration platform that automates deployment, scaling, networking, and management of containerized applications.

2) What is a Pod?

A Pod is the smallest deployable unit in Kubernetes. It can contain one or more containers sharing network and storage.

3) What is a Node?

A Node is a worker machine (VM or physical) that runs Pods.

4) What is a Cluster?

A Cluster is a set of control plane components and worker nodes managing container workloads.

5) What is a Namespace?

Namespaces logically isolate resources within a cluster.

6) What is an Init Container?

An Init Container runs before app containers and must complete successfully before main containers start.

7) What is a Deployment?

A Deployment manages ReplicaSets and ensures desired Pod replicas are running.

8) What is a ReplicaSet?

ReplicaSet ensures a specified number of Pod replicas are running.

9) What is a StatefulSet?

Used for stateful applications with stable identities and persistent storage.

10) What is a DaemonSet?

Ensures a Pod runs on every node.

11) What is a Job?

Runs a Pod until it completes successfully.

12) What is a CronJob?

Runs Jobs on a scheduled basis.

13) What is a Service?

A Service exposes a set of Pods and provides stable networking.

14) Types of Services?

ClusterIP, NodePort, LoadBalancer, ExternalName.

15) What is Ingress?

Ingress provides HTTP/HTTPS routing to Services based on host/path rules.

16) How does kubectl communicate?

kubectl talks to API Server over HTTPS using kubeconfig.

17) What is etcd?

etcd is a distributed key-value store holding cluster state.

18) What is kube-apiserver?

It exposes Kubernetes API and validates requests.

19) What is Controller Manager?

It ensures desired state matches actual state (Replica, Node, Job controllers).

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20) What is Scheduler?

Assigns Pods to nodes based on resources and constraints.

21) What is kubelet?

Agent running on nodes that manages Pods.

22) What is kube-proxy?

Handles network routing to Pods using iptables/IPVS.

23) How does Pod creation work?

User applies YAML → API Server → etcd → Controller Manager → Scheduler → kubelet → Running Pod.

24) How does external traffic reach Pods?

Internet → LoadBalancer/Ingress → Service → Pods.

25) Can internet directly hit a Service?

Only if Service type is LoadBalancer or NodePort.

26) Can LoadBalancer send traffic directly to Pods?

By default via Service. Direct-to-Pod possible using IP target mode in cloud setups.

27) What are ConfigMaps?

Used to store configuration data.

28) What are Secrets?

Used to store sensitive data like passwords and tokens.

29) What are Liveness Probes?

Check if container is alive.

30) What are Readiness Probes?

Check if container is ready to serve traffic.

31) What are Startup Probes?

Used for slow-starting applications.

32) What is HPA?

Horizontal Pod Autoscaler scales Pods based on metrics.

33) What are resource requests and limits?

Requests reserve resources; limits cap usage.

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34) What is RBAC?
Role-Based Access Control for authorization.

35) What are Taints and Tolerations?
Used to control Pod scheduling on nodes.

36) What is Node Affinity?
Rules to schedule Pods on specific nodes.

37) What is CNI?
Container Network Interface managing Pod networking.

38) What is a Persistent Volume (PV)?
Cluster-level storage resource.

39) What is Persistent Volume Claim (PVC)?
Request for storage by a Pod.

40) What is Helm?
Package manager for Kubernetes.

41) How to upgrade cluster?
Upgrade control plane first, then worker nodes using cordon and drain.

42) How to take backup?
Take etcd snapshot and use Velero for workloads and volumes.

43) What is Service Mesh?
Infrastructure layer for service-to-service communication.

44) How does Istio work?
Injects Envoy sidecars to manage traffic, security (mTLS), and observability.

45) What is mTLS?
Mutual TLS encrypts traffic between services.

46) What is Istio Gateway?
Entry/exit point for mesh traffic.

47) What is VirtualService?
Defines routing rules in Istio.

48) What is DestinationRule?
Defines policies like subsets and load balancing.

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49) Can Service A call Service B?

Yes, using internal DNS and ClusterIP.

50) Can Service B call Service C?

Yes, via application-level communication.

51) What is Rolling Update?

Gradually replaces old Pods with new ones.

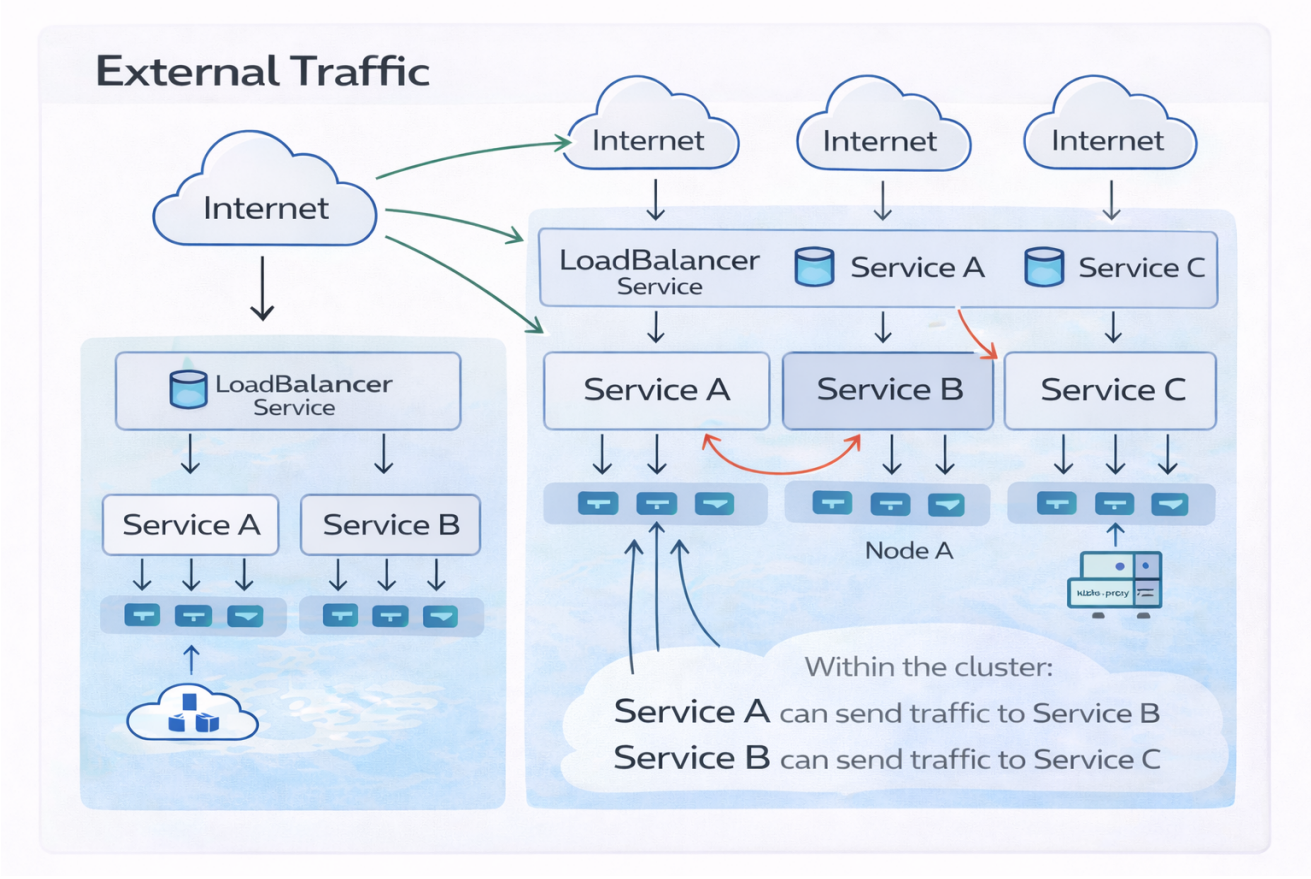
52) What is Blue-Green deployment?

Switch traffic between two environments.

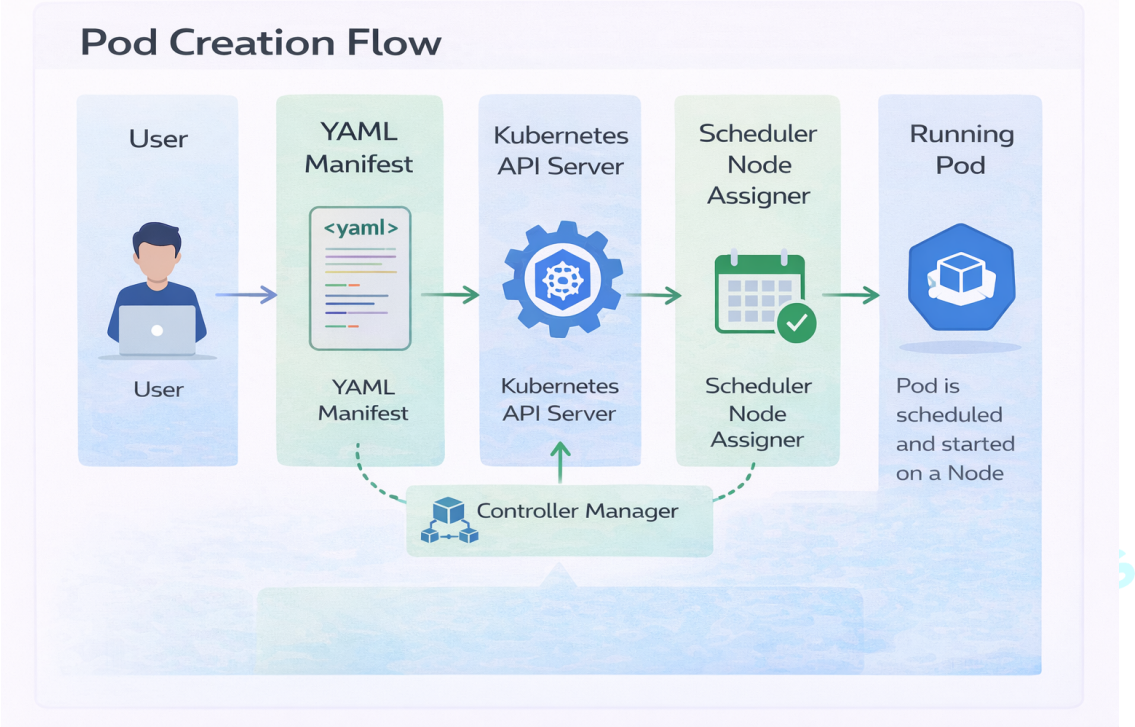
53) What is Canary deployment?

Gradually roll out new version to subset of users.

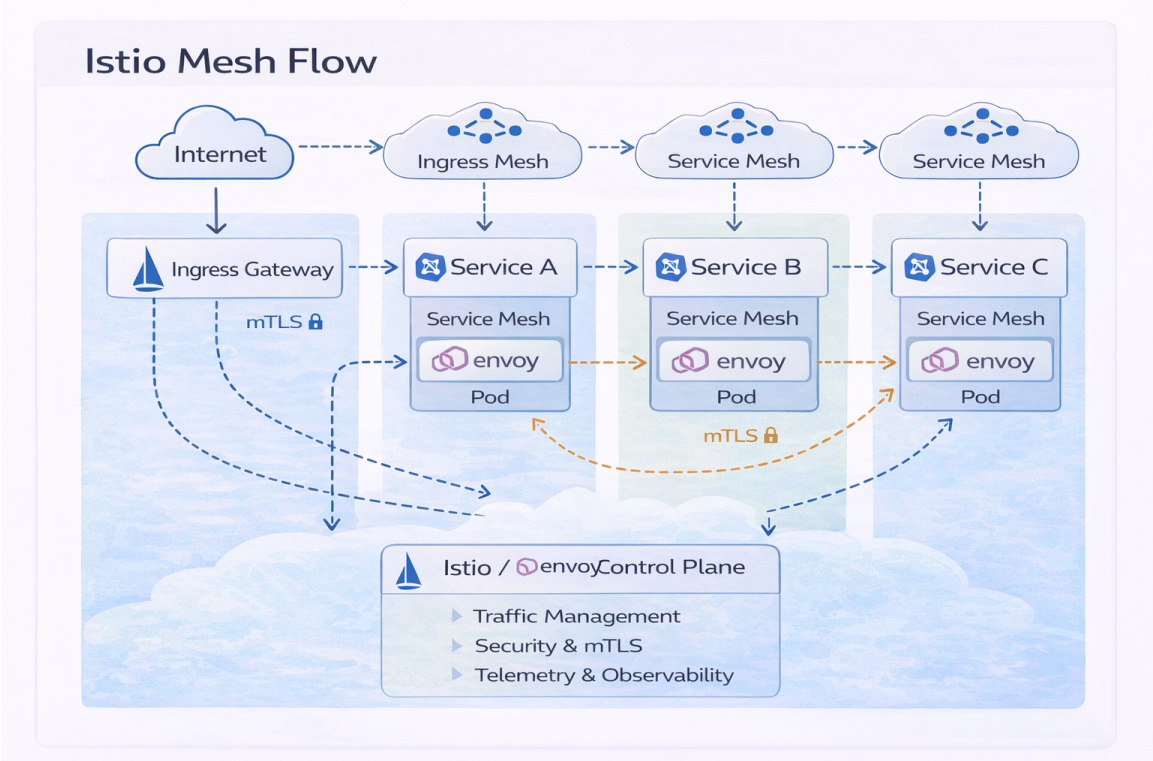
External Traffic Flow Diagram



Pod Creation Flow Diagram



Istio Mesh Flow Diagram



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Advanced Scenario-Based Interview Questions (4–8 Years Experience)

1) Pod is stuck in Pending state. What will you check?

Check node resources (CPU/Memory), taints & tolerations, node affinity rules, PVC binding issues, and scheduler events using kubectl describe pod.

2) Pods are running but Service is not routing traffic. Why?

Check Service selector labels match Pod labels, endpoints exist, readiness probe status, and kube-proxy networking rules.

3) Application latency suddenly increased after scaling. Why?

Possible causes: resource throttling, HPA misconfiguration, network bottleneck, uneven traffic distribution, or cold start delays.

4) Node NotReady issue. How do you troubleshoot?

Check kubelet status, node conditions, disk pressure, memory pressure, network connectivity, and container runtime logs.

5) How would you design multi-tenant Kubernetes architecture?

Use namespaces, RBAC isolation, network policies, resource quotas, and separate node pools if required.

6) How to implement zero-downtime deployment?

Use Rolling Updates with proper readiness probes, maxUnavailable=0, and maxSurge configuration.

7) How to secure Kubernetes cluster?

Enable RBAC, use Network Policies, enable mTLS (Istio), restrict API access, rotate secrets, and use OPA/Gatekeeper policies.

8) How do you debug CrashLoopBackOff?

Check container logs, describe pod events, verify environment variables, resource limits, and startup probe settings.

9) How would you migrate a monolith to microservices in Kubernetes?

Break services gradually, containerize, define APIs, implement CI/CD, use Ingress or service mesh for routing.

10) What happens if etcd goes down?

Cluster state becomes unavailable. Control plane stops functioning. Restore using etcd snapshot backup.

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11) How to scale cluster automatically?

Use Cluster Autoscaler integrated with cloud provider to add/remove worker nodes.

12) Difference between HPA, VPA, and Cluster Autoscaler?

HPA scales Pods horizontally, VPA adjusts Pod resource limits, Cluster Autoscaler scales nodes.

13) How to troubleshoot networking issues between Pods?

Check CNI plugin status, network policies, DNS resolution, service endpoints, and pod-to-pod connectivity.

14) How to implement canary deployment in Kubernetes?

Use separate deployments and adjust traffic weights via Ingress or Istio VirtualService.

15) How to monitor Kubernetes in production?

Use Prometheus, Grafana, ELK stack, distributed tracing (Jaeger), and alerting systems.

16) How to handle secret management in enterprise?

Use external secret managers like AWS Secrets Manager or HashiCorp Vault.

17) How to ensure high availability of control plane?

Run multiple control plane nodes with etcd cluster and load balancer in front of API server.

18) How to reduce container image pull time?

Use smaller base images, multi-stage builds, image caching, and private registry near cluster.

19) What is Pod Disruption Budget (PDB)?

Ensures minimum number of Pods remain available during voluntary disruptions.

20) How to design disaster recovery strategy for Kubernetes?

Regular etcd backups, PV snapshots, multi-region setup, GitOps configuration backup, and tested restore procedures.

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Enterprise-Level & Real Production Scenario Discussions

Production Incident Case Study 1 – High API Latency After Deployment

Imagine you deployed a new version of an application and suddenly latency increased across all services.

As a senior engineer, you would first validate whether the issue is application-related or infrastructure-related. You would check resource utilization (CPU throttling, memory pressure), HPA events, node health, and network latency. You would inspect Prometheus metrics, compare old vs new container images, verify readiness/liveness probes, and confirm no misconfiguration in ConfigMaps or Secrets. Rollback strategy and traffic shifting (canary) should also be discussed in such a scenario.

Production Incident Case Study 2 – Cluster Node Suddenly NotReady

Suppose one worker node goes into NotReady state during peak traffic.
A senior engineer should immediately check kubelet logs, disk pressure, memory pressure, and container runtime health. You should verify whether Cluster Autoscaler replaced the node, whether Pods were rescheduled successfully, and whether Pod Disruption Budgets prevented movement. Root cause analysis and prevention steps must be explained.

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Whiteboard Architecture Design – Design a Highly Available Kubernetes Setup

If asked to design a production-ready Kubernetes architecture, you should propose multi-AZ worker nodes, highly available control plane, external load balancer in front of API servers, etcd cluster replication, separate node groups for workloads, autoscaling, centralized logging, monitoring stack, network policies, and disaster recovery strategy including backups and multi-region failover.

Whiteboard Architecture – Multi-Tenant SaaS Platform Design

Explain how you would isolate multiple customers in the same cluster. Use namespaces, RBAC policies, resource quotas, network policies, separate ingress rules, and optionally separate node pools for high-security tenants. Discuss cost optimization and observability separation.

EKS Specific Interview Discussion

In EKS, control plane is managed by AWS. Worker nodes can be EC2 or Fargate. Traffic typically flows via ALB/NLB using AWS Load Balancer Controller. IAM Roles for Service Accounts (IRSA) is used for secure AWS access. Cluster Autoscaler integrates with Auto Scaling Groups.

AKS Specific Interview Discussion

In AKS, Azure manages the control plane. Integration with Azure AD provides authentication. Azure Load Balancer and Application Gateway are commonly used. You should discuss node pools, availability zones, and Azure Monitor integration.

GKE Specific Interview Discussion

In GKE, Google manages control plane. Node pools can be zonal or regional. GKE Autopilot abstracts node management. Cloud Load Balancer integrates natively and Workload Identity secures GCP access.

Kubernetes + DevOps CI/CD Integration Scenario

Explain how Kubernetes integrates with CI/CD pipelines. Code is built via Jenkins/GitLab CI, containerized using Docker, pushed to registry, and deployed via Helm or GitOps (ArgoCD/Flux). You should discuss versioning, rollback strategy, environment promotion, and security scanning.

Zero Downtime Production Upgrade Strategy

Describe rolling updates with maxSurge and maxUnavailable settings. Explain readiness probes importance, database backward compatibility, and traffic splitting using Istio or Ingress-based canary deployment.

Disaster Recovery & Backup Strategy in Enterprise

A strong answer should include etcd snapshot backups, persistent volume snapshots, infrastructure as code backup, GitOps configuration recovery, and regular restore testing. Multi-region active-passive or active-active strategies should also be explained.

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